higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

WELDING
NQF LEVEL 4

(6031034)

12 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

1.1 Choose a description from COLUMN B that matches a/an item/word in COLUMN A. Write only the letter (A–L) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.1.1	Arc Blow	A	bevel angle on the metal edge
1.1.2	Chamfer	B	external or internal force applied on an object
1.1.3	Flux	C	the presents of gas pockets or voids in metal
1.1.4	Chamfer	D	when an arc wanders or whip from its normal course during arc welding
1.1.5	Stress	E	an electrode holder
1.1.6	Neutral flame	F	the bevelling the edge of a plate at an angle for welding
1.1.7	Porosity	G	a chemical used to promote fusion during welding
1.1.8	Stinger	H	metals containing no iron
1.1.9	Ultimate strength	I	weld joint
1.1.10	Non-ferrous	J	the maximum strain a metal is capable of withstanding
		K	flux coating
		L	produced when an equal amount of oxygen and fuel gas is burnt

(10 × 1) (10)

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1.2 FIGURE 1 below shows a shielded metal arc welding (SMAW) set up. Label Items (A–F) in your ANSWER BOOK.

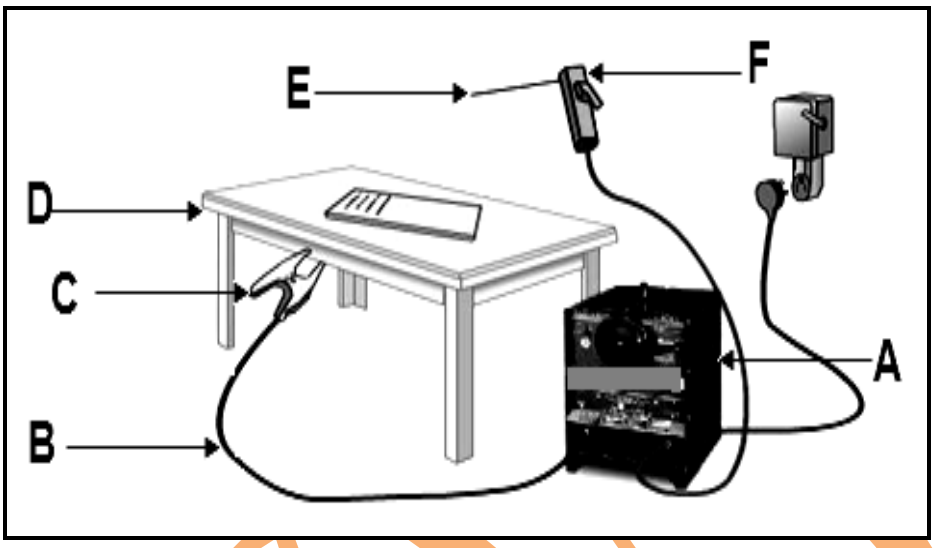


FIGURE 1 (6)

1.3 It is important to correctly align pipes before welding can take place.

With the aid of a neat, labelled drawing, draw a set-up showing the use of an angle iron stand to align pipes. (4)
[20]

QUESTION 2

2.1 Name the FOUR welding positions used for the welding of pipes using shielded metal arc welding (SMAW) equipment. (4)

2.2 Name FIVE good characteristics of an electrode used for shielded metal arc welding (SMAW). (5)

2.3 Shielded metal arc welding (SMAW) is the most common method used to weld pipes together.

Explain the shielded metal arc welding (SMAW) process. (6)

2.4 Name FIVE welding defects that can occur when welding pipes together. (5)
[20]

QUESTION 3

3.1 Name SIX components that make up a gas tungsten arc welding (GTAW) set up. (6)

3.2 Complete the following paragraph by using the word(s) in the list below. Write only the word(s) next to the question number (3.2.1–3.2.5) in the ANSWER BOOK.

odourless; inert gas welding; non-consumable; contamination; inert gas

Gas tungsten arc welding (GTAW) is a welding process that is also called (3.2.1). The welding process uses a (3.2.2) tungsten electrode. (3.2.3), referred to the shielding gas, protects the welding from atmospheric (3.2.4). The shielding gas is transparent and (3.2.5) does not affect the welders' working conditions. (5)

3.3 Explain 'direct current electrode positive (DCEP)'. (4)

3.4 Name FIVE safety precautions that must be adhered to when transporting gas cylinders. (5)

[20]

QUESTION 4

4.1 FIGURE 2 shows a gas tungsten arc welding torch.
Label items (A–F) in your ANSWER BOOK.

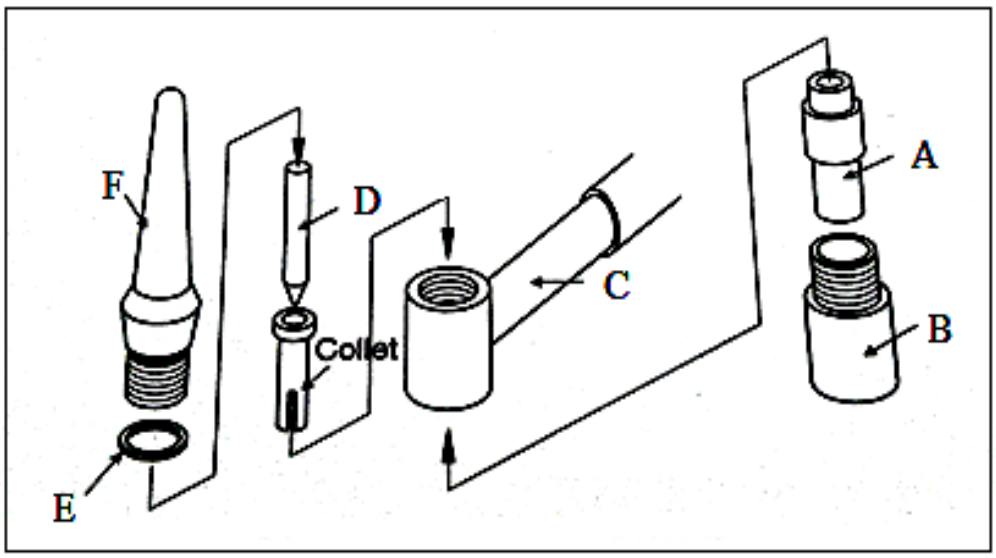


FIGURE 2 (6)

4.2 Name FIVE advantages of gas tungsten arc welding. (5)

4.3 Explain why a tungsten electrode is classified as a 'non-consumable' item. (4)

4.4 Name FIVE pre-checks that must be adhered to when preparing gas tungsten arc welding equipment for the welding process. (5)
[20]

QUESTION 5

5.1 Name SIX safety precautions that a welder must adhere to when performing metal inert gas (MIG) welding. (6)

5.2 Explain what radiographic welding inspections are. (4)

5.3 Name FIVE disadvantages of metal inert gas welding (MIG). (5)

5.4 Name FIVE causes of spatter when using the metal inert gas welding (MIG) process. (5)
[20]

TOTAL: 100

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